ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO4 – INVESTIGATIVE ASSIGNMENT

ULTRA-HIGH-SPEED PHOTONIC TRANSPORT LINK

Technology Evaluation, Impairment Mitigation, and Integrated System Design

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Abstract

This report investigates a 32-channel DWDM optical transport system carrying 200 Gb/s per channel, for a gross aggregate capacity of 6.4 Tb/s. It compares modulation formats, photonic modulator mechanisms, optical amplifiers, WDM impairments and coherent digital signal processing (DSP), then proposes an integrated architecture. The supplied assignment omits numerical per-channel input/output powers, fiber span length, an exact OSNR/FEC target and simulator outputs. These missing values are identified rather than represented as measured results.

Stage 1 – Interpret the Communication Requirement

Capacity and design baseline

Aggregate capacity = 32 × 200 Gb/s = 6,400 Gb/s = 6.4 Tb/s

The system must provide high spectral efficiency, adequate OSNR, acceptable pre-FEC BER/Q-factor, controlled per-channel launch power and scalability. The assignment names ITU-T G.698, IEEE 802.3ct/ck and OIF-CEI. Exact compliance must be checked against the selected application code and the latest applicable interface specifications.

Metric	Purpose
Spectral efficiency (b/s/Hz)	Measures transmitted data per unit optical bandwidth.
OSNR	Signal power relative to ASE noise, stated with a reference bandwidth.
BER and Q-factor	Measure decision quality and error performance.
Launch power	Balances receiver power against Kerr nonlinearities.
Modulator bandwidth / Vπ	Must support the symbol rate with acceptable drive voltage and loss.
Thermal stability / footprint	Affects PIC integration, power, reliability and cost.

Impairment	Effect	Mitigation
Chromatic dispersion (CD)	Wavelength-dependent delay broadens pulses	Coherent DSP frequency-domain/FIR equalizer
PMD	Polarization states experience different delays	Low-PMD fiber and adaptive 2×2 MIMO equalizer
SPM / XPM	Kerr effect creates intensity-dependent phase shifts	Launch-power optimization; optional digital backpropagation
FWM	WDM channels mix into new frequencies	Grid, dispersion and power planning
ASE noise	Amplifier noise accumulates through spans	Low-noise amplifiers and OSNR monitoring
Crosstalk / PDL	Filtering/coupling and polarization-dependent loss degrade channels	Qualified AWG/MMI components, isolation and alignment

Investigation Requirement Statement: Design a scalable 32-channel DWDM link at 200 Gb/s per channel, with coherent modulation, high-speed electro-optic transmitters, controlled optical amplification and gain flatness, and coherent DSP for CD, PMD and nonlinear impairment mitigation. Validate the design using analytical calculations and simulation of OSNR, BER/Q-factor and launch power.

Stage 2 – Modulation and Modulator Selection

Format	Bits per symbol	Advantages / limitations
NRZ/RZ-OOK	1	Simple and economical, but low spectral efficiency at very high rates.
PAM4	2	Lower baud rate than OOK; requires accurate amplitude decisions and has tighter noise margin.
DPSK / DQPSK	1 / 2	Differential phase detection; intermediate complexity and efficiency.
16-QAM	4	High spectral efficiency but needs higher OSNR and linearity.
Dual-polarization 16-QAM	8 per dual-polarization symbol	High efficiency, suitable for coherent systems; requires complex coherent DSP.

Assume 200 Gb/s is gross rate before FEC. DP-16QAM carries 4 bits per symbol on each of two polarizations:

DP-16QAM baud rate = 200 Gb/s ÷ (4 × 2) = 25 GBd

PAM4 carries 2 bits per symbol. For a single-stream 200 Gb/s PAM4 comparison:

PAM4 baud rate = 200 Gb/s ÷ 2 = 100 GBd

These are not identical architectures: DP-16QAM is dual-polarization coherent, while this PAM4 estimate is a single intensity-modulated stream. If 20% FEC overhead is assumed, line rate becomes 240 Gb/s, giving 30 GBd DP-16QAM or 120 GBd PAM4.

The assignment does not specify exact grid spacing. As a design assumption, evaluate 75 GHz for 25 GBd DP-16QAM, subject to roll-off, guard band and the applicable G.698 application code.

Nominal gross spectral efficiency = 200 Gb/s ÷ 75 GHz ≈ 2.67 bit/s/Hz

A 100 GBd PAM4 stream is not a straightforward fit in a 75 GHz channel. It would need wider spacing or a different lane/subcarrier architecture.

Modulation mechanisms and architectures

Technology	Assessment
Electro-optic effect	Very fast; preferred for high-speed data modulation. Vπ and bandwidth depend on material and device design.
Acousto-optic	Acoustic transit limits speed; more suitable for switching/filtering than the main 200 Gb/s modulator.
Thermo-optic	Slow thermal response; useful for tuning/stabilization rather than symbol-by-symbol modulation.
Magneto-optic	Specialized polarization/phase applications; not the primary data modulator.
Kerr effect	Fast intensity-dependent refractive index; important in nonlinear propagation and selected all-optical functions.
MZI modulator	Broadband design possible; larger footprint but suitable for high-speed I/Q modulation.
Micro-ring resonator	Compact and wavelength-selective, but resonance is thermally sensitive and bandwidth depends on Q.

Vπ is the voltage needed to create a π phase shift. The PDF supplies no device-specific Vπ or bandwidth, so numeric values must come from the selected manufacturer datasheet or simulation. Silicon photonics offers dense integration; lithium niobate is attractive for low-loss, high-linearity electro-optic modulation.

Design Decision 1: I recommend dual-polarization coherent 16-QAM using an electro-optic I/Q Mach–Zehnder modulator. It carries 8 bits per dual-polarization symbol and needs approximately 25 GBd for 200 Gb/s gross, improving spectral efficiency compared with the 100 GBd single-stream PAM4 estimate. Its higher OSNR requirement and DSP complexity must be verified.

Stage 3 – Optical Amplification Configuration

Amplifier	Strengths	Limitations
EDFA	Mature C-band multi-channel gain and good output power	ASE accumulates; gain tilt and saturation need control.
SOA	Compact and integrable	Noise, crosstalk and nonlinear effects can limit long-haul use.
Lumped Raman	Can improve effective noise performance	Pump complexity and safety/control needs.

Distributed Raman	Gain distributed along the transmission fiber	Pump lasers and careful engineering are required.
Hybrid Raman + EDFA	Combines Raman noise benefits with EDFA output gain	More complex, costly and power-consuming; needs optimization.

Power calculation (input values missing in source)

The assignment's amplifier problem states that per-channel input and target output powers should be assumed, but the numerical values are blank in the supplied PDF. A unique numerical gain cannot therefore be calculated from the source.

Total output power (dBm) = P_channel,out (dBm) + 10 $\log_{10}$(32)

10 $\log_{10}$(32) = 15.05 dB

Total output power (W) = 10^((P_total,dBm − 30)/10)

Span gain (dB) = P_channel,out − P_channel,in

Illustrative example only: if output is −3 dBm/channel, total output is 12.05 dBm ≈ 16.0 mW. If input is −15 dBm/channel, required gain is 12 dB. These are assumed examples, not given assignment values.

EDFA gain is not perfectly flat across the C-band. A gain-flattening filter (GFF) and dynamic per-channel equalization reduce channel power differences. ASE accumulates through cascaded amplifiers, so span loss, noise figure, gain and amplifier count must be jointly optimized.

Design Decision 2: I recommend hybrid distributed Raman + EDFA amplification with gain flattening and per-channel equalization. It can balance noise performance and output power for high-capacity WDM, but deployment approval depends on filling in the missing power values and verifying OSNR across all spans.

Stage 4 – Impairment Compensation and DSP

Method / impairment	Engineering analysis
Chromatic dispersion	Different wavelengths travel at different group velocities. Coherent DSP uses frequency-domain or FIR equalization.
PMD	Polarization modes have different delays. Use low-PMD fiber and adaptive 2×2 MIMO polarization equalization.
SPM / XPM	Kerr nonlinearity causes own-channel and cross-channel phase changes. Optimize launch power; DBP may partially compensate.
FWM	WDM channels mix into new frequencies. Manage channel grid, dispersion and launch power.
PDL	Loss varies with polarization. Use low-PDL components and polarization-aware processing.
DCF	Optically compensates dispersion but adds insertion loss and can increase cost/nonlinear interactions.
Optical phase conjugation	May compensate some dispersion/nonlinear effects under suitable symmetric conditions; requires specialized design.
Coherent DSP	Flexible CD/PMD compensation without DCF insertion loss, but requires high-speed ADCs and computation.

Recommended coherent DSP sequence

- Optical front end / intradyne coherent receiver (ICR) converts the optical field to electrical I/Q signals.
- Analog conditioning and ADC sampling.
- Deskew and normalize the sampled streams.
- Chromatic-dispersion compensation using frequency-domain equalization or a long FIR filter.
- Timing recovery and adaptive 2×2 MIMO polarization demultiplexing (CMA initialization followed by a suitable decision-directed algorithm).
- Frequency-offset estimation and carrier-phase recovery.
- Symbol decisions, soft-decision demapping and FEC decoding.
- Optional digital backpropagation (DBP) using split-step Fourier propagation reversal if simulations show enough benefit to justify computation and power.

Design Decision 3: I recommend coherent DSP with CD equalization, adaptive 2×2 MIMO polarization recovery, carrier recovery, demapping and FEC. DBP should be optional and included only if simulation shows useful Q/BER improvement. This avoids the insertion loss of DCF and supports a scalable coherent WDM architecture.

Stage 5 – Integrated DWDM Strategy

Proposed architecture: 32 narrow-linewidth tunable lasers feed 32 dual-polarization coherent transmitters. Each generates a 200 Gb/s DP-16QAM channel. An AWG multiplexer combines channels; isolation and launch-power control feed single-mode fiber spans. Hybrid Raman–EDFA amplification with GFF maintains channel power. An AWG demultiplexer separates the channels, each of which feeds a coherent ICR, ADC and DSP/FEC chain.

Block diagram:

32 tunable lasers → DP-16QAM EO I/Q MZI transmitters → channel monitoring → AWG multiplexer → isolator / launch-power control → single-mode fiber span(s) + distributed Raman pump(s) → EDFA + gain-flattening filter → further spans as required → AWG demultiplexer → coherent ICR → ADC → CD equalizer → adaptive 2×2 MIMO/PMD equalizer → carrier recovery → demapper → FEC decoder → 200 Gb/s recovered channel

Parameter	Proposed design	Status
Channel count / rate	32 × 200 Gb/s	Given
Aggregate capacity	6.4 Tb/s gross	Calculated
Modulation	DP-16QAM	Selected
Symbol rate	25 GBd gross; increase for FEC overhead	Calculated under stated assumption
Grid spacing	Evaluate 75 GHz	Assumption; validate spectral occupancy and G.698 code
Laser linewidth	Use qualified coherent transceiver specification	Not provided
Modulator	EO I/Q MZI, silicon photonics or lithium niobate	Architecture selected; datasheet needed
Vπ / bandwidth	Device-specific	Not provided; must validate
Amplifier	Hybrid Raman–EDFA + GFF + channel equalization	Selected; numeric gain plan pending
DSP	CD, MIMO/PMD, timing/carrier recovery, demapping, FEC; optional DBP	Selected

DSP filter taps	Determine from sampling rate, dispersion, PMD and implementation limits	Simulation/implementation dependent

Key trade-offs: DP-16QAM improves spectral efficiency but demands better OSNR and more accurate phase/amplitude recovery. AWG/MMI insertion loss reduces launch power. Excess launch power increases SPM/XPM/FWM, while insufficient power reduces receiver OSNR. Gain tilt can make the worst channel fail even when average channel performance looks acceptable.

Design Decision 4: I recommend the integrated coherent DP-16QAM, AWG, hybrid Raman–EDFA and DSP architecture as the preferred candidate for 6.4 Tb/s. It is not yet deployment-ready until span loss, gain, OSNR, BER/Q-factor and standard application-code compliance are confirmed.

Stage 6 – Verification Table

Requirement	Target	Evidence / proposed result	Status
Aggregate capacity	6.4 Tb/s	32 × 200 Gb/s = 6.4 Tb/s	Meets arithmetic target
Modulation efficiency	DP-16QAM	8 bits/dual-polarization symbol; 25 GBd gross	Consistent; OSNR validation needed
Grid spacing	ITU-T G.698 grid	75 GHz considered	Not confirmed; validate application code
Modulator bandwidth/footprint	200 Gb/s channel, compact PIC	EO I/Q MZI selected	Device data needed
Amplifier gain/flatness	Meet power budget and flat C-band gain	Hybrid Raman–EDFA + GFF proposed	Numeric validation pending
OSNR / BER	Meet transceiver/FEC thresholds	Assignment gives no numerical threshold	Cannot claim verified
DSP	CD/PMD and nonlinear tolerance	CD equalizer, MIMO, carrier recovery, FEC; optional DBP	Architecture selected; simulate
Standards	G.698 and applicable IEEE/OIF specs	Compliance checklist proposed	Exact interface code needed

Simulation Plan (OptiSystem / OptSim / MATLAB)

The assignment requests eye diagrams, optical spectra, constellation plots and Q-factor versus launch-power curves. No simulator outputs or complete link parameters were supplied, so this report does not fabricate plots or claim simulation has been completed.

- Model 32 channels using the chosen grid and 200 Gb/s DP-16QAM transceivers.
- Enter actual per-channel launch power, span length, fiber attenuation/dispersion/PMD/nonlinear coefficient, and component insertion losses.
- Model AWG MUX/DEMUX, amplifier gain/noise figure, gain tilt, GFF and Raman pump conditions.
- Compare baseline DSP with CD/PMD compensation and optional DBP.

- Sweep launch power (e.g. −5 to +3 dBm as a test range only, not a recommended operating range) and record pre-FEC BER, Q-factor and OSNR.
- Export optical spectra, constellation and eye diagrams at representative channels; compare EDFA-only against hybrid Raman–EDFA.
- Report the worst channel, not only the center channel, and repeat under component/temperature tolerances.

Final Engineering Decision – Would You Deploy?

NO — not directly from the present data. The proposed architecture is technically promising, but the assignment omits essential numerical inputs: per-channel input/target output powers for the amplifier calculation, fiber length/span loss, exact grid/application code, OSNR/FEC target and simulation results. Therefore the full power budget, amplifier gain, OSNR margin, pre-FEC BER and Q-factor cannot be verified.

Redesign / completion actions:

- Confirm whether 200 Gb/s is gross or net of FEC and select the actual transceiver and line rate.
- Specify carrier grid and verify the exact G.698 application code and applicable IEEE/OIF interface.
- Obtain intended per-channel input/output powers, span lengths, fiber loss, amplifier noise figures and gain limits; calculate the complete 32-channel power budget.
- Model hybrid Raman–EDFA gain and ASE accumulation, add GFF/equalization and verify worst-channel OSNR margin.
- Run OptiSystem/OptSim/MATLAB and include actual spectra, constellations, eye diagrams, Q/BER and launch-power sweep.
- Verify modulator bandwidth/$V\pi$, laser linewidth, PIC loss, thermal stability, DSP power, reliability and cost from datasheets.

Deployment can be reconsidered after the missing data are supplied and the worst-case channel meets the selected transceiver's OSNR and pre-FEC BER limits with engineering margin.

Conclusion

The proposed 32-channel system offers 6.4 Tb/s gross capacity. DP-16QAM carries 8 bits per dual-polarization symbol and requires 25 GBd at 200 Gb/s gross under the stated assumption, compared with 100 GBd for single-stream PAM4. An EO I/Q MZI transmitter, AWG multiplexing, hybrid Raman–EDFA amplification with gain flattening and coherent DSP form a coherent integrated architecture. CD, PMD, nonlinear effects, ASE and crosstalk must be addressed together. The correct engineering decision is not to deploy until the missing power parameters are supplied and simulation validates OSNR, BER/Q-factor, spectral occupancy and launch-power optimization.

References and Source Basis

Primary source: ECA18 – Optoelectronics and Optical Communication, CO4 Investigative Assignment PDF supplied with the task. The report follows its six-stage structure. General standards named by the assignment: ITU-T G.698 family; IEEE 802.3ct/802.3ck; OIF-CEI. Consult the current official standards and selected transceiver/component datasheets before claiming compliance.